UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	:	**INFORMATION SYSTEMS ENGINEERING**
COURSE CODE	:	**ISP550/ITS572**
EXAMINATION	:	**FEBRUARY 2025**
TIME	:	**3 HOURS**

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of 2 parts: PART A (10 Questions)
   PART B (4 Questions)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of
   i) the Question Paper
   ii) an Answer Booklet – provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

PART A

State whether each of the following statements is **TRUE** or **FALSE.**

1. Systems analysis is sometimes referred to as "understanding and specification."
2. Reuse is commonly applied to two different development technologies: Object-oriented development and component-based development.
3. One important step in estimating the time required for a project is to try to identify all of the functional requirements of the new system.
4. Business case is a major outcome and deliverable from the project initiation and planning phase that contains the best estimate of a project's scope, benefits, costs, risks, and resource requirements.
5. If the analysts understand the major business processes, it is not usually necessary to create a comprehensive list of all business processes.
6. In a use case diagram, an actor must always be a person.
7. The output of the design activities is a set of diagrams and documents that describe the solution system.
8. Designing the application architecture involves specifying in detail how some system activities will be carried out.
9. The greater the number of customers the greater the maintenance costs.
10. Configuration management is a process of ensuring that only authorized changes are made to the system.

(10 marks)

PART B

QUESTION 1

a) Describe **TWO** (2) differences between data and information within the context of Information Systems.

(4 marks)

b) List the **SIX** (6) different sources of software.

(6 marks)

c) Off-the-shelf software refers to pre-packaged software solutions that are developed for a wide range of users, rather than being customized for a specific organization. Briefly explain **TWO** (2) factors to consider when evaluating off-the-shelf software for an organization.

(6 marks)

d) Briefly explain **TWO** (2) issues related to understanding during the requirements elicitation.

(4 marks)

QUESTION 2

a) The initiation phase of an information systems project involves assessing the size, scope, and complexity of the project and establishing the necessary procedures to support subsequent phases.

List **FOUR** (4) key activities involved in the initiation phase of an information systems project.

(4 marks)

b) Key reasons for project failure in information systems development include poor planning, unclear requirements, lack of stakeholder engagement, ineffective communication, inadequate risk management, and insufficient resource allocation.

Explain how effective project management practices can prevent these failures.

(6 marks)

c) Project feasibility contributes to ensuring the project is viable from all angles, helping stakeholders make informed decisions about whether to proceed with the project.

Briefly explain the **THREE** (3) different categories of project feasibility that need to be assessed.

(6 marks)

d) Explain **THREE** (3) key deliverables of the system implementation process to minimize the chances of errors and increase user satisfaction.

(6 marks)

QUESTION 3

a) Explain how the CRUD (Create, Read, Update and Delete)technique validates and refine use cases in requirements analysis.

(4 marks)

b) Create a use case diagram for a university online registration system based on the following narrative.

The online university registration system should enable the staff of each academic department to examine the courses offered by their department, add and remove courses, and change the information about them (e.g., the maximum number of students permitted). It should permit students to examine currently available courses, add and drop courses to and from their schedules, and examine the courses for which they are enrolled. Department staff should be able to print a variety of reports about the courses and the students enrolled in them. The system should ensure that no student registers more than four courses and that students who have any unpaid fees are not permitted to register. (Assume that a fees data store is maintained by the university's financial office, which the registration system accesses but does not change).

(8 marks)

c) Explain the difference between architectural design and detailed design in the context of information system design.

(6 marks)

d) Controls are mechanisms and procedures that are built into a system to safeguard the system and the information within it.

Describe **TWO** (2) potential consequences of neglecting security and integrity controls in designing a web-based system.

(6 marks)

QUESTION 4

a) State **FOUR** (4) key stakeholders involved in the requirements elicitation process.

(4 marks)

b) Differentiate between open and closed interviews in requirements elicitation. Provide example of how each type of interview can be used during system development.

(6 marks)

c) Metro Car Service Agency needs a system that keeps car service records. The company's system analyst has provided information about the problem domain in the form of notes. Your job is to use those notes to draw the domain model class diagram. The analyst's notes are as follows:

- The Owner class has attributes name and address.
- The Vehicle class is an abstract class that has attributes registration number, model, and model year.
- The Manufacturer class has attributes name and location.
- The Dealer class has attributes name and address.

A service record is an association class between each vehicle and a dealer, with attributes, service date and current mileage. A warranty service record is a special type of service record with an additional attribute: eligibility verification. Each service record is associated with a predefined service type, with attributes type ID, description, and labor cost. Each service type is associated with zero or more parts, with attributes part ID, description, and unit cost. Parts are used with one or more service types. An owner can own many vehicles, and a vehicle can be owned by many owners. An owner and a vehicle are entered into the system only when an owned vehicle is first serviced by a dealer. Vehicles are serviced many times at various dealers, which service many vehicles.

Draw UML domain model class diagram for the system as described. Be as specific and accurate as possible, given the information provided. If needed information is not given, make realistic assumptions.

(8 marks)

d) Maintenance requires a systematic and structured approach to implement changes to the information system and ensure their quality and compatibility.

Explain the difference between corrective maintenance and adaptive maintenance in system support.

(6 marks)

END OF QUESTION PAPER

